

$$\left[\begin{array}{cc|c} 1 & k & 1 \\ k & 1 & -1 \end{array} \right] \xrightarrow{2) \quad \begin{array}{l} R_2 \rightarrow R_2 - kR_1 \\ R_1 \rightarrow R_1 - kR_2 \end{array}} \left[\begin{array}{cc|c} 1 & k & 1 \\ 0 & 1-k^2 & -1-k \end{array} \right]$$

rank = 1 for no sol or ∞ sol

$$2) \quad 1-k^2=0$$

$$\boxed{k = \pm 1}$$

Substituting in original Equation, $\Rightarrow k=1$ gives no solⁿ

$k=-1$ gives ∞ solⁿ

if $k \neq \pm 1 \Rightarrow \text{Rank} = 2 \Rightarrow \text{Unique solution,}$